Hashem Alomar

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Education

University of Maryland, College Park **May 2027**

Combined Bachelor & Master of Science in Computer Science and Mathematics *GPA: 3.9*

Relevant Coursework: Algorithms, Data Structures, Discrete Mathematics, Real Analysis, Statistics, Probability Theory, Linear Algebra, Multivariable Calculus, Object-Oriented Programming, Competitive Programming, Financial Mathematics

Experience

University of Maryland **August 2026 – Present**

Independent Researcher

- Fine-tuned Qwen3-8B (LoRA SFT, 240 examples) to control what its chain of thought is about. This raised CoTControl's held-out ignore_question score from **0% to 88–96%** under the benchmark's official grader.
- Built CoT-necessity monitors for Qwen3-8B (4,374 questions, 7 task families); a linear probe stayed robust to a fine-tuned sandbagger that broke behavioral checks (AUROC 0.93→0.49 vs 0.78→0.77).
- Showed that distilling another model's rewritten reasoning makes Qwen3-8B's CoT stop reporting a leaked answer key (**98%→4%**) while still using it; replicated across seeds, model sizes, a real teacher and 9 hint types.

Stripe **May 2026 – August 2026**

Software Engineering Intern

- Design an end-to-end real-time validation system for change event streams, enabling detection of **10,00,000+** missing or incorrect events per month in O(minutes).
- Architect a change event aggregator in Go that ingests write-ahead logs at **100k RPS** peak throughput.
- Design a battle-tested alert system and run book to help engineers debug missing data in under **20 minutes**.

University of Maryland **January 2025 – May 2026**

Research Assistant - Quantum Machine Learning

- Optimized QLSTM circuitry using **PennyLane**, to predict 10+ equities concurrently, **saving ∼$100 per iteration.**
- Achieved **7% higher model accuracy** by testing QLSTM models on **IonQ** hardware across 3–8 qubits.
- **Reduced RMSE (error) by 12%** by optimizing QLSTM circuit depth and topology, leveraging IonQ API.

GeneticFit Labs **September 2024 – March 2025**

Chief Technology Officer

- Founded and served as CTO of a genetic training startup that raised $150K in seed funding, with a **$3M valuation.**
- Built and deployed an iOS MVP in **Swift and Firebase**, tested by 20+ pilot users**, with a 95% satisfaction rate.**
- **Led a 10+ team** of software engineers and designers from ideation through full-stack deployment.

Projects & Extracurriculars

Maryland AI Safety Initiative — *Fellow* **August 2026 – Present**

- Selected Fellow in MASI, a competitive AI Safety fellowship at UMD; studied AI governance and risk while leading technical reading groups on alignment and interpretability

ARENA Education **August 2026 – October 2026**

- Completed ARENA's alignment research engineering curriculum covering transformer internals, mech interp (TransformerLens, SAEs), and RLHF.

ProjectLIFT — *Founder* **April 2025 – Present**

- Founded a student-run club delivering tech solutions to 10+ local small businesses, saving over $200,000+ in expenses.

Math Club — *Officer* **August 2024 – Present**

- Leading weekly problem-solving sessions and discussions across diverse mathematical topics in club with 100+ students.
- Tutoring 100+ students in Calculus, Linear Algebra, Statistics and Differential Equations.

Skills, Awards & Interest

Programs: Assembly, Python, Java, C, C++, SQL, Pytorch, NumPy, Pandas, TensorFlow, Scikit Learn, PennyLane, Go
Awards: Bloomberg BPuzzled 2nd Place, USA Climbing Divisionals: 2nd Place, Jordanian Karate National Champion(Youth)
Interests: Poker, Rock Climbing, Hiking, Puzzles, Blackjack, Chicago Bulls, Philosophy